

Agricultural price trends in India

Introduction

- Agriculture plays a key role in India's economy. It has a wide range of crops, livestock, etc. Knowing the Price trend of agriculture commodity is very important for farmers, traders, policymakers, researchers, economist, consumers. As it helps to better understand and make the best decisions out of it, through price trends. The future production of certain crops can also be predicted. According to the price trend Traders do their supply and there is demand in the market, For policymakers certain sustainability farming policies can be suggested by looking at the price trend, reducing carbon footprint and fossil fuel consumption, economist can better predict the inflation or over supply of the commodity in the market. Below are the Data visualization through python which will help better understand the data through various plots.

Description of the dataset

Importing Libraries
Importing and reading the txt file
reading the data frame
describing the shape
counting the values in the dataframe.

Objective

- To analyse the given dataset, which is price trends in agriculture commodity. And how it affects the various sections of the society, who are interested in the agriculture commodities. for eg:- farmers, traders, policymakers, economist, Researchers.

Problem Definitions

- To know the price trends at different levels, minimum, maximum, modal and commodities in the market.

Observations

- The dataset has detailed information regarding the agricultural commodities of state and districts in india
- There were variables as such: State and district, Market and commodities and varieties of commodities, arrival details of the commodities in the market, extensive varieties of agricultural commodities were there like grains and nuts etc.
- The analysis has been conducted for price trend analysis, commodities sold in the market, commodities sold in the market with the maximum price, minimum price and modal price. (No. of commodities in each state, district wise commodity distribution of chhattisgarh, correlation of min, max, modal price catplots, and scatterplot.)

Results

- Price trends analysis
- Commodities related with other variables
- State and district wise distributed commodities
- Economic Assessment

Recommendations & Conclusion

- Conclusions: Data-driven decisions are mostly accurate. Through this datasets and it's visualizations, Insights can be taken to make an impactful decisions that are profitable in nature, There is wide varieties of commodities of data has been taken. There are interdependences among the variable, and variations in price.
- Recommendations: For the traders, analysis can be done for the market-timing, supply chain optimization for the commodities, taking the variable (arrival date), which can actually help many of the traders in supply and logistics.

References

- Used geek for geeks, w3 school, Chatgpt & sample codes from google colab

Steps

1.Create a Jupyter Notebook File with name in the following format - REGNO_Lab1.ipynb 2.Import the given dataset (text file) into Python 3.Do the basic illustration of the features and data as available in the Dataset 4.Provide statistical measures, inter-dependency (correlation) between variables, trends, summaries, etc. 5.Create Visualisations through Seaborn Libraries, and interpret

In [343]:  *#Importing the required Libraries into the notebook*

```
import pandas as pd
import warnings
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt
warnings.filterwarnings('ignore')
```

In [344]:  *#reading the txt file into csv file*

```
df = pd.read_csv("C:\\Users\\sd\\Downloads\\SHREYA KISPOTTA 22122317 - datase
```

In [345]:  *#dataframe*

```
df
```

Out[345]:

	state	district	market	commodity	variety	arrival_date	min_price	r
0	Andaman and Nicobar	South Andaman	Port Blair	Amaranthus	Amaranthus	04-03-2019	6000	
1	Kerala	Ernakulam	Piravam	Amaranthus	Other	04-03-2019	1500	
2	Kerala	Kasargod	Kasargod	Amaranthus	Other	04-03-2019	1500	
3	Kerala	Kasargod	Manjeswaram	Amaranthus	Amaranthus	04-03-2019	2400	
4	Kerala	Kollam	Anchal	Amaranthus	Other	04-03-2019	4500	
...	...	...	...	...	...	...	...	
2233	Uttar Pradesh	Muzaffarnagar	Thanabhawan	Wood	Other	04-03-2019	250	
2234	Uttar Pradesh	Saharanpur	Gangoh	Wood	Beete (Rose)	04-03-2019	415	
2235	Uttar Pradesh	Saharanpur	Nakud	Wood	Other	04-03-2019	250	
2236	Uttar Pradesh	Saharanpur	Nanuta	Wood	Other	04-03-2019	350	
2237	Uttar Pradesh	Sitapur	Viswan	Wood	Other	04-03-2019	200	

2238 rows × 9 columns



```
In [346]: #defining the data-frame types  
df.dtypes
```

```
Out[346]: state          object  
district        object  
market          object  
commodity       object  
variety         object  
arrival_date    object  
min_price       int64  
max_price       int64  
modal_price     int64  
dtype: object
```

```
In [347]: #defining the data-frame shape  
df.shape
```

```
Out[347]: (2238, 9)
```

```
In [348]: #counting the data- frame values  
df.count()
```

```
Out[348]: state          2238  
district        2238  
market          2238  
commodity       2238  
variety         2238  
arrival_date    2238  
min_price       2238  
max_price       2238  
modal_price     2238  
dtype: int64
```

```
In [349]: #checking if any duplicate values are there or not  
df = df.drop_duplicates()  
df.head(5)
```

```
Out[349]:
```

	state	district	market	commodity	variety	arrival_date	min_price	max_pri
0	Andaman and Nicobar	South Andaman	Port Blair	Amaranthus	Amaranthus	04-03-2019	6000	80
1	Kerala	Ernakulam	Piravam	Amaranthus	Other	04-03-2019	1500	20
2	Kerala	Kasargod	Kasargod	Amaranthus	Other	04-03-2019	1500	16
3	Kerala	Kasargod	Manjeswaram	Amaranthus	Amaranthus	04-03-2019	2400	34
4	Kerala	Kollam	Anchal	Amaranthus	Other	04-03-2019	4500	47

```
In [350]: ▶ #defining the commodity count
df.commodity.count()
```

Out[350]: 2236

```
In [353]: ▶ #describing the detailed commodity of the data
df['commodity'].unique()
```

```
Out[353]: array(['Amaranthus', 'Amla(Nelli Kai)', 'Amphophalus', 'Apple',
'Arecanut(Betelnut/Supari)', 'Arhar (Tur/Red Gram)(Whole)',
'Arhar Dal(Tur Dal)', 'Ashgourd', 'Bajra(Pearl Millet/Cumbu)',
'Banana', 'Banana - Green', 'Barley (Jau)', 'Beans', 'Beetroot',
'Bengal Gram Dal (Chana Dal)', 'Bengal Gram(Gram)(Whole)',
'Ber(Zizyphus/Borehannu)', 'Betel Leaves', 'Bhindi(Ladies Finger)',
'Bitter gourd', 'Black Gram (Urd Beans)(Whole)',
'Black Gram Dal (Urd Dal)', 'Black pepper', 'Bottle gourd',
'Brinjal', 'Broken Rice', 'Cabbage', 'Capsicum', 'Carrot',
'Cashewnuts', 'Castor Seed', 'Cauliflower', 'Chikoos(Sapota)',
'Chilly Capsicum', 'Cluster beans', 'Coconut', 'Coconut Oil',
'Coconut Seed', 'Colacasia', 'Copra', 'Coriander(Leaves)',
'Corriander seed', 'Cotton', 'Cowpea (Lobia/Karamani)',
'Cowpea(Veg)', 'Cucumbar(Kheera)', 'Drumstick', 'Dry Chillies',
'Duster Beans', 'Elephant Yam (Suran)', 'Field Pea', 'Fish',
'French Beans (Frasbean)', 'Garlic', 'Ghee', 'Gingelly Oil',
'Ginger(Dry)', 'Ginger(Green)', 'Gram Raw(Chholia)', 'Grapes',
'Green Avare (W)', 'Green Chilli', 'Green Gram (Moong)(Whole)',
'Green Gram Dal (Moong Dal)', 'Green Peas', 'Ground Nut Seed',
'Groundnut', 'Guar', 'Guar Seed(Cluster Beans Seed)', 'Guava',
'Gur(Jaggery)', 'Hybrid Cumbu', 'Indian Beans (Seam)',
'Jowar(Sorghum)', 'Jute', 'Karamani', 'Karbuja(Musk Melon)',
'Kinnow', 'Knool Khol', 'Kodo Millet(Varagu)',
'Kulthi(Horse Gram)', 'Leafy Vegetable', 'Lemon',
'Lentil (Masur)(Whole)', 'Lime', 'Linseed',
'Little gourd (Kundru)', 'Long Melon(Kakri)', 'Maida Atta',
'Maize', 'Mango', 'Mango (Raw-Ripe)', 'Mashrooms', 'Masur Dal',
'Methi(Leaves)', 'Mint(Pudina)', 'Moath Dal',
'Mousambi(Sweet Lime)', 'Mustard', 'Mustard Oil',
'Niger Seed (Ramtil)', 'Onion', 'Onion Green', 'Orange',
'Paddy(Dhan)(Basmati)', 'Paddy(Dhan)(Common)', 'Papaya',
'Papaya (Raw)', 'Pear(Marasebu)', 'Peas cod', 'Peas Wet',
'Peas(Dry)', 'Pegeon Pea (Arhar Fali)', 'Pepper garbled',
'Pineapple', 'Plum', 'Pointed gourd (Parval)', 'Pomegranate',
'Potato', 'Pumpkin', 'Raddish', 'Ragi (Finger Millet)', 'Rajgir',
'Rice', 'Ridgeguard(Tori)', 'Round gourd', 'Rubber',
'Seemebadnekai', '"Sesamum(Sesame,Gingelly,Til)"', 'Snakeguard',
'Soyabean', 'Spinach', 'Sponge gourd', 'Squash(Chappal Kadoo)',
'Sugar', 'Surat Beans (Papadi)', 'Suvarna Gadde', 'Sweet Potato',
'Sweet Pumpkin', 'T.V. Cumbu', 'Tamarind Fruit', 'Tapioca',
'Taramira', 'Tender Coconut', 'Thinai (Italian Millet)', 'Tinda',
'Tobacco', 'Tomato', 'Turmeric', 'Turmeric (raw)', 'Turnip',
'Water Melon', 'Wheat', 'Wheat Atta', 'White Pumpkin', 'Wood'],
dtype=object)
```

```
In [354]: #describing the detailed district of the data  
df['district'].unique()
```

```
Out[354]: array(['South Andaman', 'Ernakulam', 'Kasargod', 'Kollam', 'Malappuram',  
                'Thirssur', 'Thiruvananthapuram', 'Alappuzha', 'Jalandhar',  
                'Faridabad', 'Mewat', 'Kangra', 'Dhar', 'Kolhapur', 'Pune',  
                'Satara', 'Muktsar', 'Tarntaran', 'Agra', 'Baghpat', 'Bahraich',  
                'Bhadohi(Sant Ravi Nagar)', 'Ghaziabad', 'Kannuj', 'Kanpur',  
                'Lakhimpur', 'Mirzapur', 'Muradabad', 'Muzaffarnagar',  
                'Saharanpur', 'Dehradun', 'Sonitpur', 'North Goa', 'Kannur',  
                'East Khasi Hills', 'Mokokchung', 'Panchmahals', 'Nagpur',  
                'Nanded', 'Chitrakut', 'Burdwan', 'Mau(Maunathbhanjan)',  
                'Sonbhadra', 'Puruliya', 'Kottayam', 'Anand', 'Sikar', 'Tonk',  
                'Aligarh', 'Bulandshahar', 'Etah', 'Etawah', 'Ambala', 'Gurgaon',  
                'Kullu', 'Jalgaon', 'Sholapur', 'Bishnupur', 'Chandel',  
                'Imphal East', 'Imphal West', 'Thoubal', 'Gajapati', 'Mansa',  
                'Jalore', 'Ballia', 'Dhenkanal', 'Sundergarh', 'Bangalore',  
                'Kolar', 'Mayurbhanja', 'Kurnool', 'Kota', 'Mahoba', 'Amreli',  
                'Bharuch', 'Surat', 'Vadodara(Baroda)', 'Badwani', 'Jhabua',  
                'Darrang', 'Kamrup', 'Valsad', 'Sepahijala', 'Jalpaiguri',  
                'North 24 Parganas', 'South 24 Parganas', 'Dhubri', 'Cuddalore',  
                'Vellore', 'Villupuram', 'Jhansi', 'Angul', 'Balasore', 'Ganjam',  
                'Gurdaspur', 'Hoshiarpur', 'Patiala', 'Ranga Reddy',  
                'Gautam Budh Nagar', 'Shahjahanpur', 'Bankura', 'Nadia',  
                'Nowrangpur', 'Narsinghpur', 'Ludhiana', 'Sheopur', 'Amritsar',  
                'Barnala', 'Ropar (Rupnagar)', 'Nagercoil (Kanniyakumari)',  
                'Panipat', 'Badgam', 'Coimbatore', 'Dindigul', 'Erode', 'Madurai',  
                'Namakkal', 'Salem', 'Sivaganga', 'Dharmapuri', 'Krishnagiri',  
                'Tumkur', 'Karaikal', 'Buldhana', 'Karimnagar', 'Bastar',  
                'Hyderabad', 'Jorhat', 'Surajpur', 'Thiruvannamalai', 'Bikaner',  
                'Kachchh', 'Barmer', 'Allahabad', 'Fatehpur', 'Khiri (Lakhimpur)',  
                'Sitapur', 'Medinipur(W)', 'Malda', 'Murshidabad',  
                'Uttar Dinajpur', 'Cachar', 'Dindori', 'Farukhabad',  
                'North Tripura', 'Baran', 'Jhalawar', 'Rampur', 'Ahmednagar',  
                'Nashik', 'Jharsuguda', 'Bhatinda', 'South District',  
                'UdhamSinghNagar', 'West Godavari', 'Kanker', 'Anupur', 'Bargarh',  
                'Kancheepuram', 'Ramanathapuram', 'Thanjavur', 'Virudhunagar',  
                'Nalgonda', 'Nizamabad', 'Raebareilly', 'Kheda', 'Kuruksheetra',  
                'Hooghly', 'Chittor', 'Khandwa', 'Chittorgarh', 'Jalaun (Orai)',  
                'Khammam'], dtype=object)
```

```
In [355]: #describing the detailed state of the data  
df['state'].unique()
```

```
Out[355]: array(['Andaman and Nicobar', 'Kerala', 'Punjab', 'Haryana',  
                'Himachal Pradesh', 'Madhya Pradesh', 'Maharashtra',  
                'Uttar Pradesh', 'Uttarakhand', 'Assam', 'Goa', 'Meghalaya',  
                'Nagaland', 'Gujarat', 'West Bengal', 'Rajasthan', 'Manipur',  
                'Odisha', 'Karnataka', 'Andhra Pradesh', 'Tripura', 'Tamil Nadu',  
                'Telangana', 'Jammu and Kashmir', 'Pondicherry', 'Chattisgarh'],  
                dtype=object)
```

```
In [357]: #dropping the null value in the data-frame  
print(df.isnull().sum())
```

```
state          0  
district       0  
market         0  
commodity      0  
variety        0  
arrival_date   0  
min_price      0  
max_price      0  
modal_price    0  
dtype: int64
```

```
In [358]: #Describe the data  
df.describe()
```

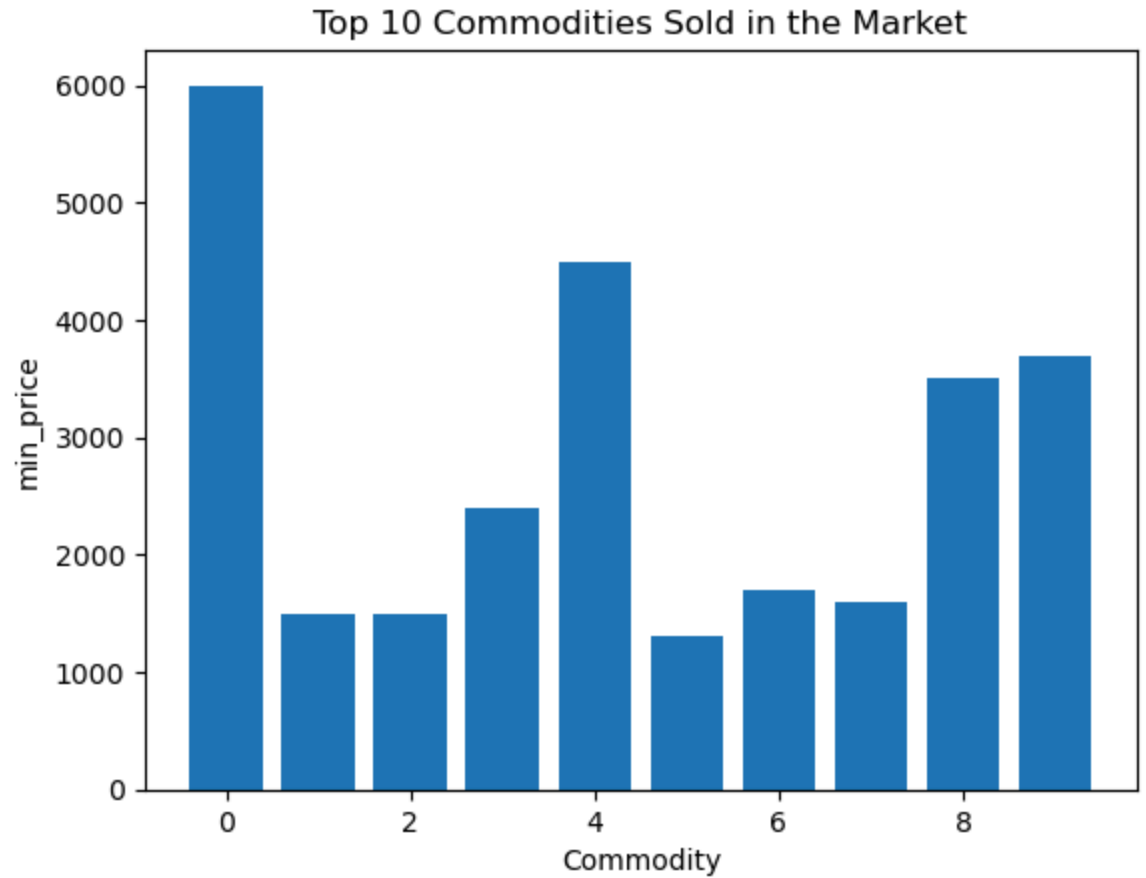
Out[358]:

	min_price	max_price	modal_price
count	2236.000000	2236.000000	2236.000000
mean	2451.271914	2851.289356	2694.124329
std	3450.055096	3915.978829	3693.077586
min	0.000000	0.000000	2.000000
25%	800.000000	1000.000000	950.000000
50%	1700.000000	1957.500000	1830.000000
75%	3200.000000	3700.000000	3500.000000
max	110000.000000	130000.000000	120000.000000

```
In [362]: # Calculate the Pearson correlation coefficient for each pair of variables  
corr_min_max = np.corrcoef(df['min_price'], df['max_price'])[0, 1]  
corr_min_modal = np.corrcoef(df['min_price'], df['modal_price'])[0,1]
```

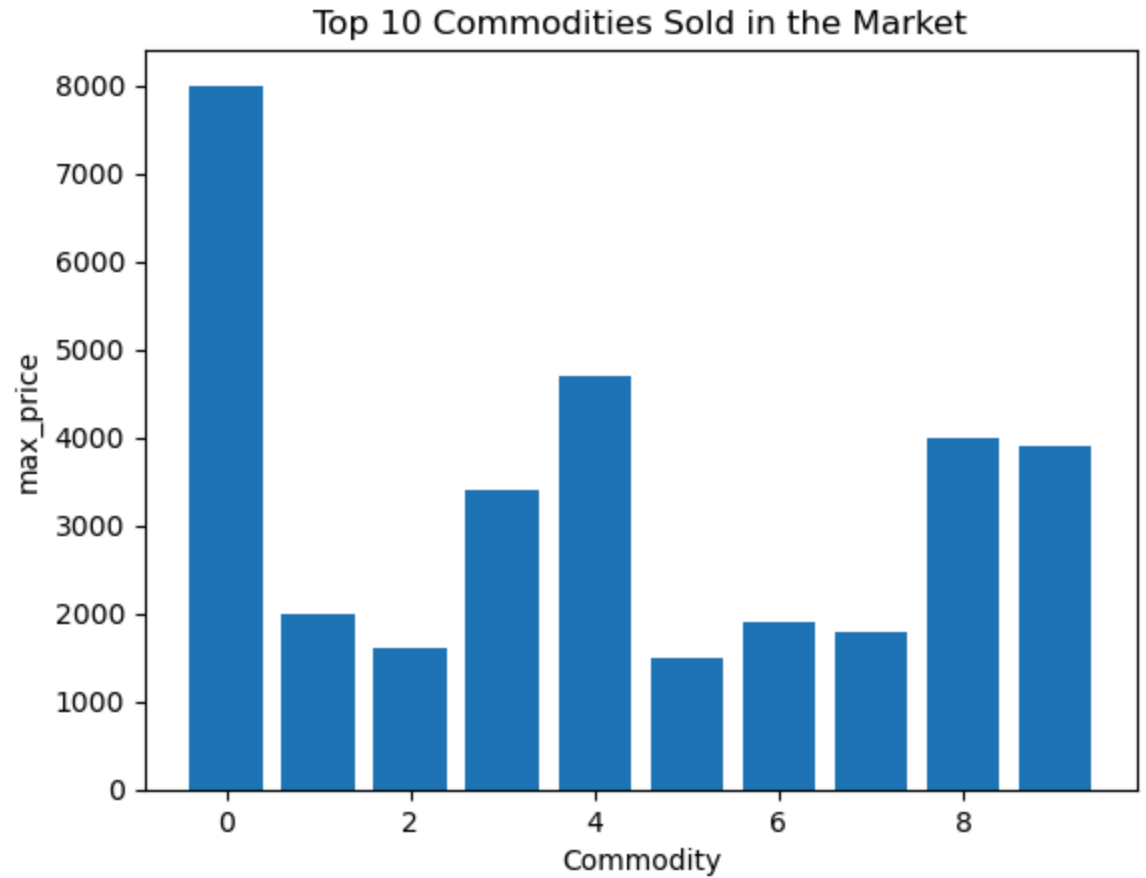
In [363]: `# Line plot of the top 10 commodities sold in the market with the minimum price`

```
plt.bar(df.index[:10], df['min_price'][:10])  
plt.xlabel('Commodity')  
plt.ylabel('min_price')  
plt.title('Top 10 Commodities Sold in the Market')  
plt.show()
```



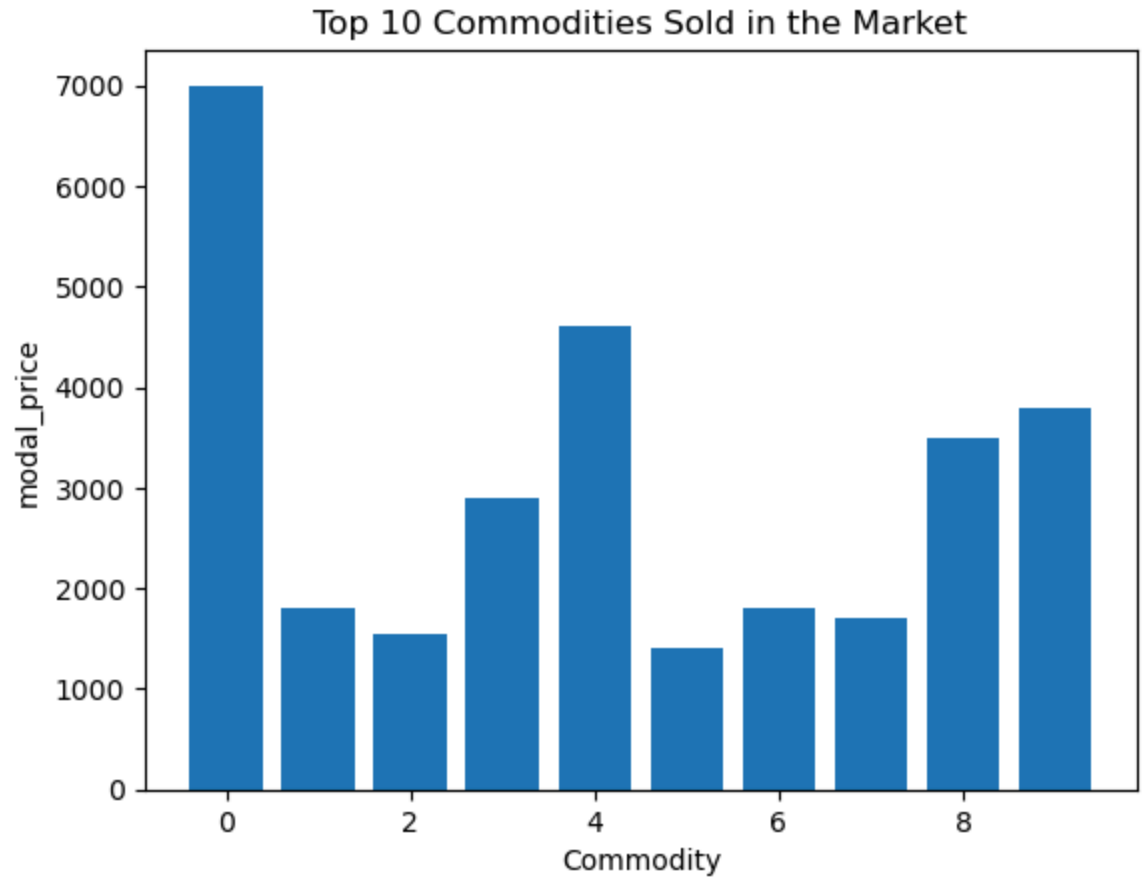
In [364]: `# Line plot of the top 10 commodities sold in the market with the maximum price`

```
plt.bar(df.index[:10], df['max_price'][:10])  
plt.xlabel('Commodity')  
plt.ylabel('max_price')  
plt.title('Top 10 Commodities Sold in the Market')  
plt.show()
```

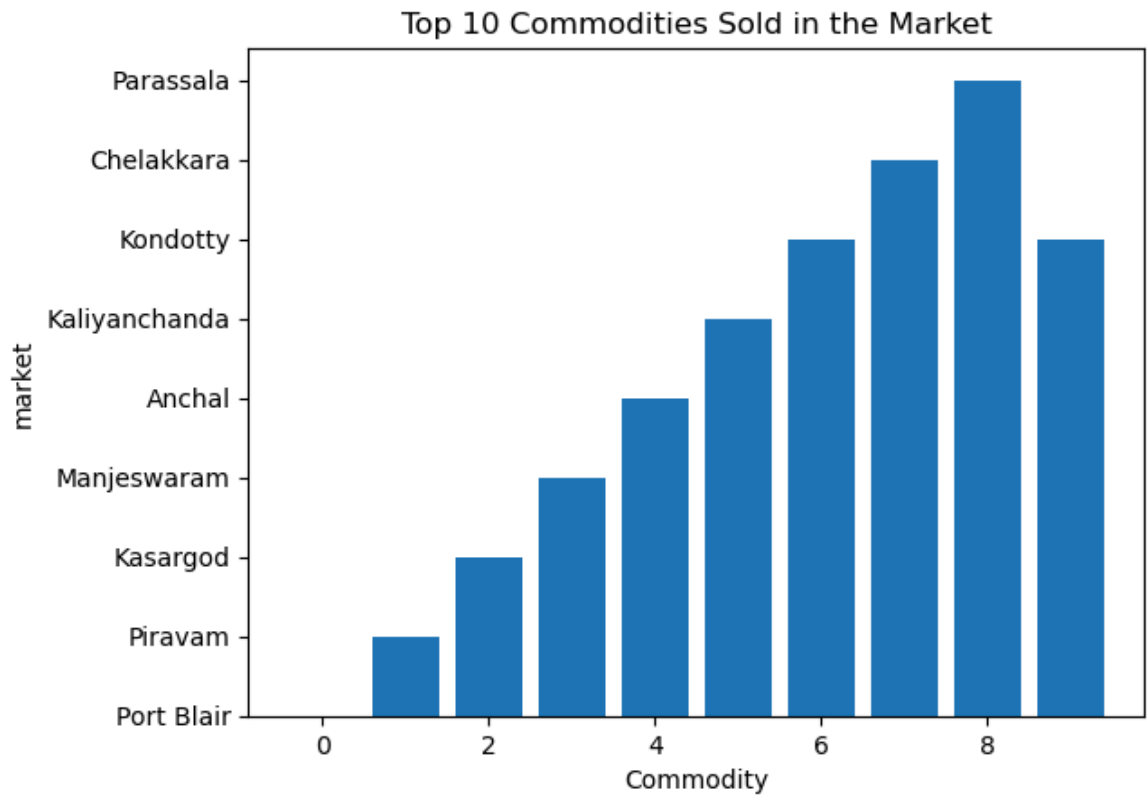


In [367]: `# Line plot of the top 10 commodities sold in the market with the modal price`

```
plt.bar(df.index[:10], df['modal_price'][:10])  
plt.xlabel('Commodity')  
plt.ylabel('modal_price')  
plt.title('Top 10 Commodities Sold in the Market')  
plt.show()
```



```
In [368]: # a line plot of the top 10 commodities sold in the market  
# code generated with AI  
plt.bar(df.index[:10], df['market'][:10])  
plt.xlabel('Commodity')  
plt.ylabel('market')  
plt.title('Top 10 Commodities Sold in the Market')  
plt.show()
```



```
In [369]: #defining correlation between the minimum prices, maximum prices, modal price  
df.corr()
```

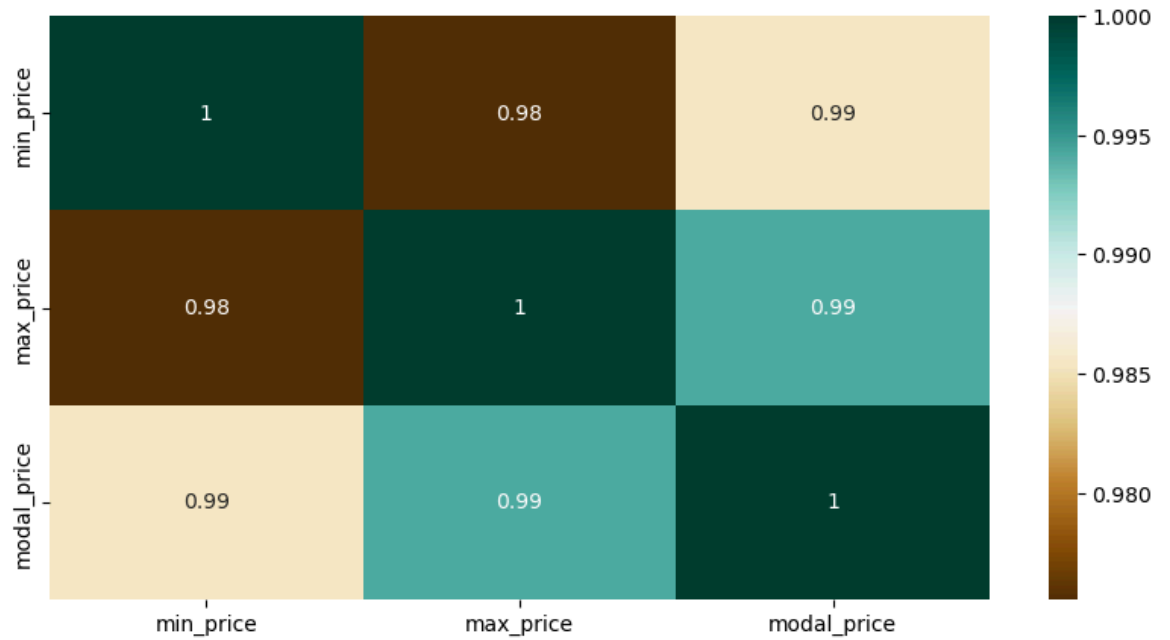
Out[369]:

	min_price	max_price	modal_price
min_price	1.000000	0.975517	0.985323
max_price	0.975517	1.000000	0.994257
modal_price	0.985323	0.994257	1.000000

```
In [370]: ▶ #Showing correlation of minimum prices, maximum prices, modal price through h
plt.figure(figsize=(10,5))
c= df.corr()
sns.heatmap(c,cmap="BrBG",annot=True)
c
```

Out[370]:

	min_price	max_price	modal_price
min_price	1.000000	0.975517	0.985323
max_price	0.975517	1.000000	0.994257
modal_price	0.985323	0.994257	1.000000



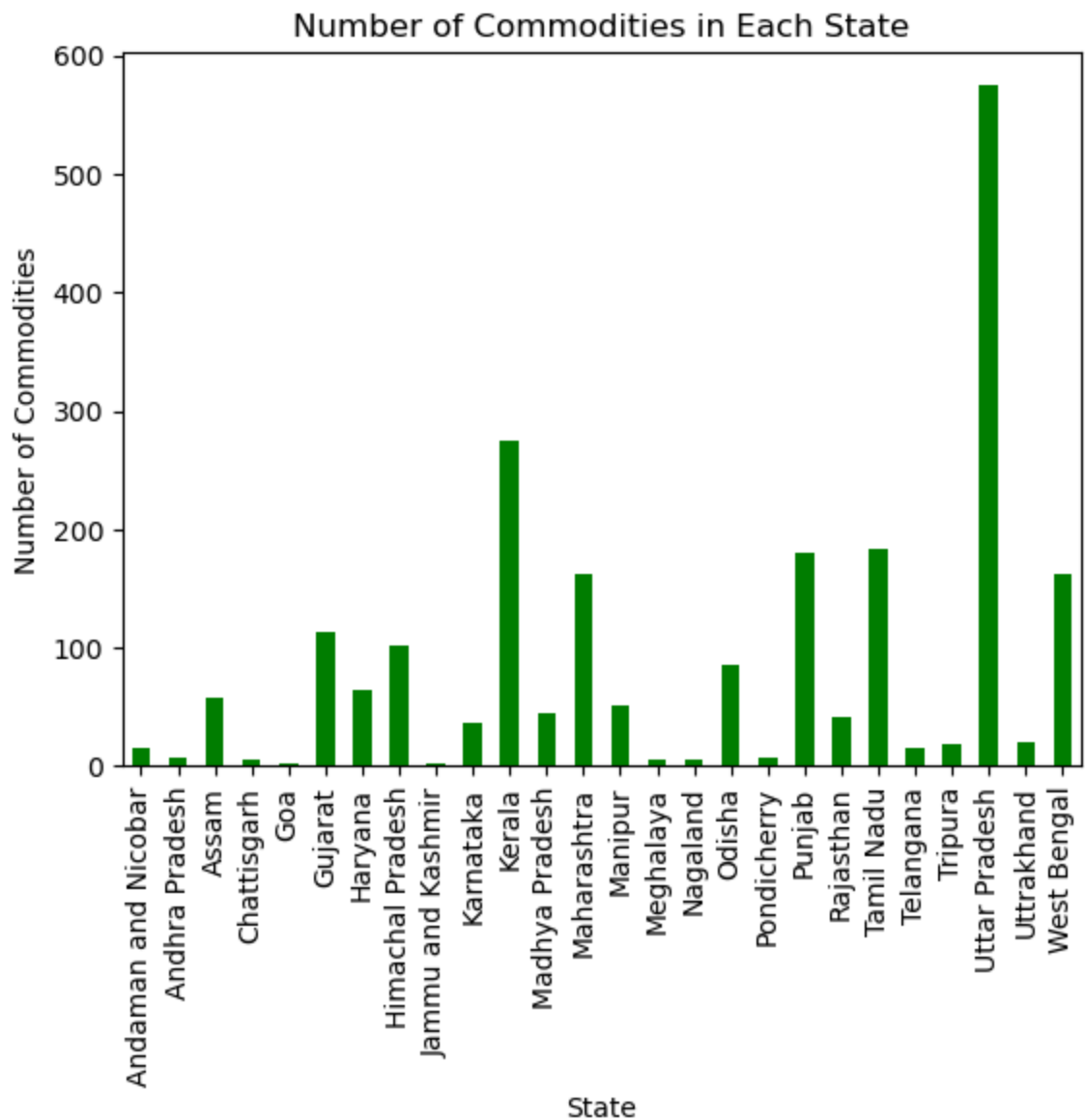
```
In [371]: ▶ #defining the number of commodities into each states

df = pd.read_csv("C:\\Users\\sd\\Downloads\\SHREYA KISPOTTA 22122317 - dataset\\data.csv")

# Create a bar chart of the number of commodities in each state
df.groupby('state').size().plot(kind='bar', color = 'green')

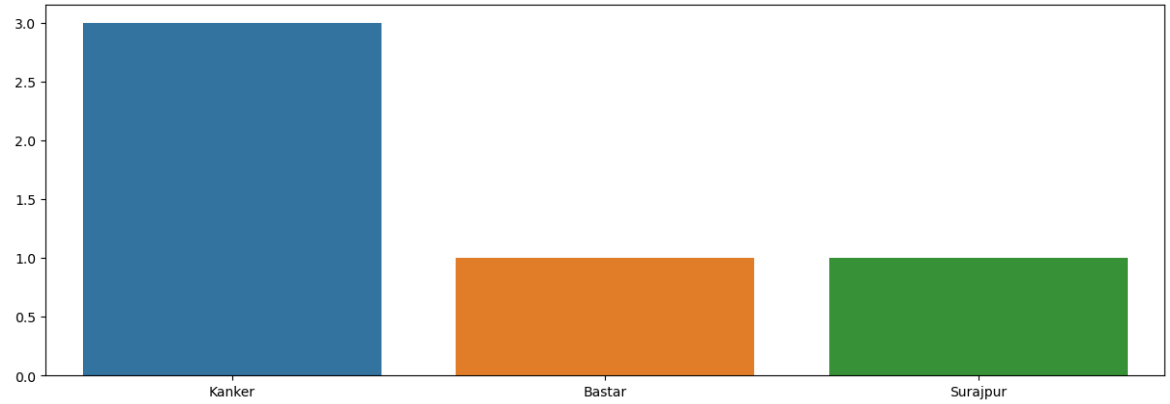
# Add a title and labels to the chart
plt.title('Number of Commodities in Each State')
plt.xlabel('State')
plt.ylabel('Number of Commodities')

# Show the chart
plt.show()
```



```
In [372]: ▶ #showing the commodities, district wise of chhattisgarh through bar plots
values = df[df.state == 'Chattisgarh'].district.value_counts().to_list()[ :10]
labels = df[df.state == 'Chattisgarh'].district.value_counts().index[:10]

plt.figure(figsize=(15, 5))
sns.barplot(x=labels, y=values)
plt.show()
```



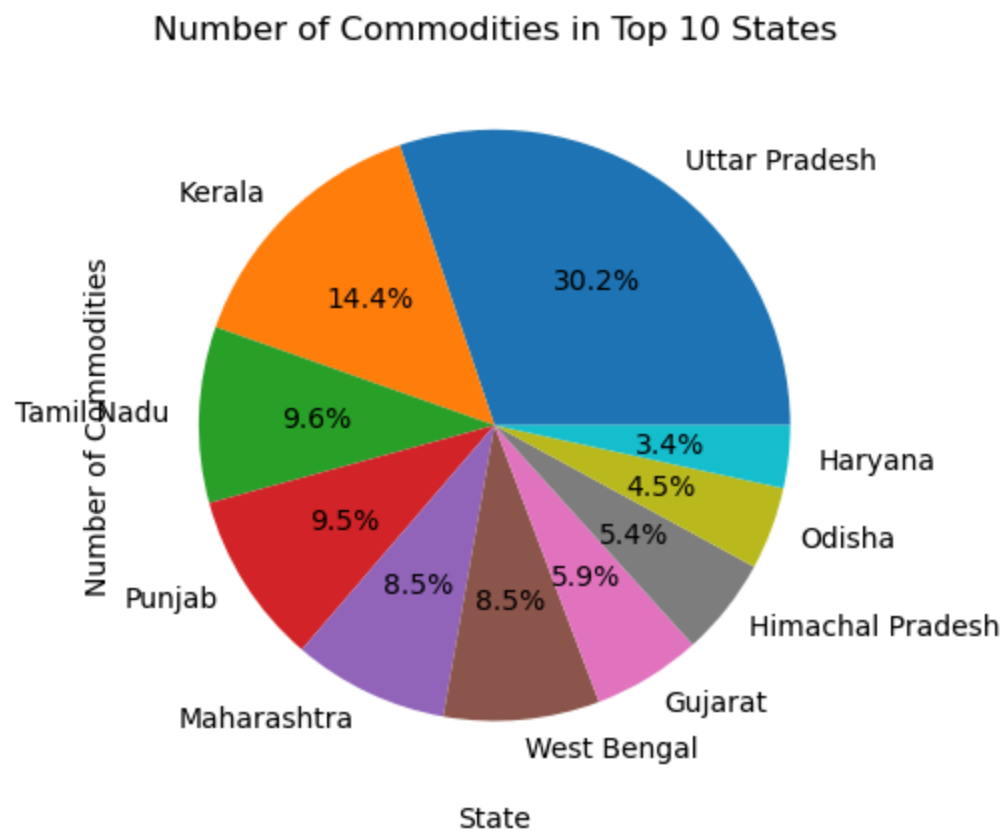
```
In [373]: #showing the number of commodities into the states.

df = pd.read_csv("C:\\Users\\sd\\Downloads\\SHREYA KISPOTTA 22122317 - dataset\\commodities.csv")

# Create a pie chart of the number of commodities in each state
top_ten_states = df.groupby('state').size().nlargest(10)
top_ten_states.plot(kind='pie', autopct='%1.1f%%')

# Add a title and labels to the chart
plt.title('Number of Commodities in Top 10 States')
plt.xlabel('State')
plt.ylabel('Number of Commodities')

# Show the chart
plt.show()
```

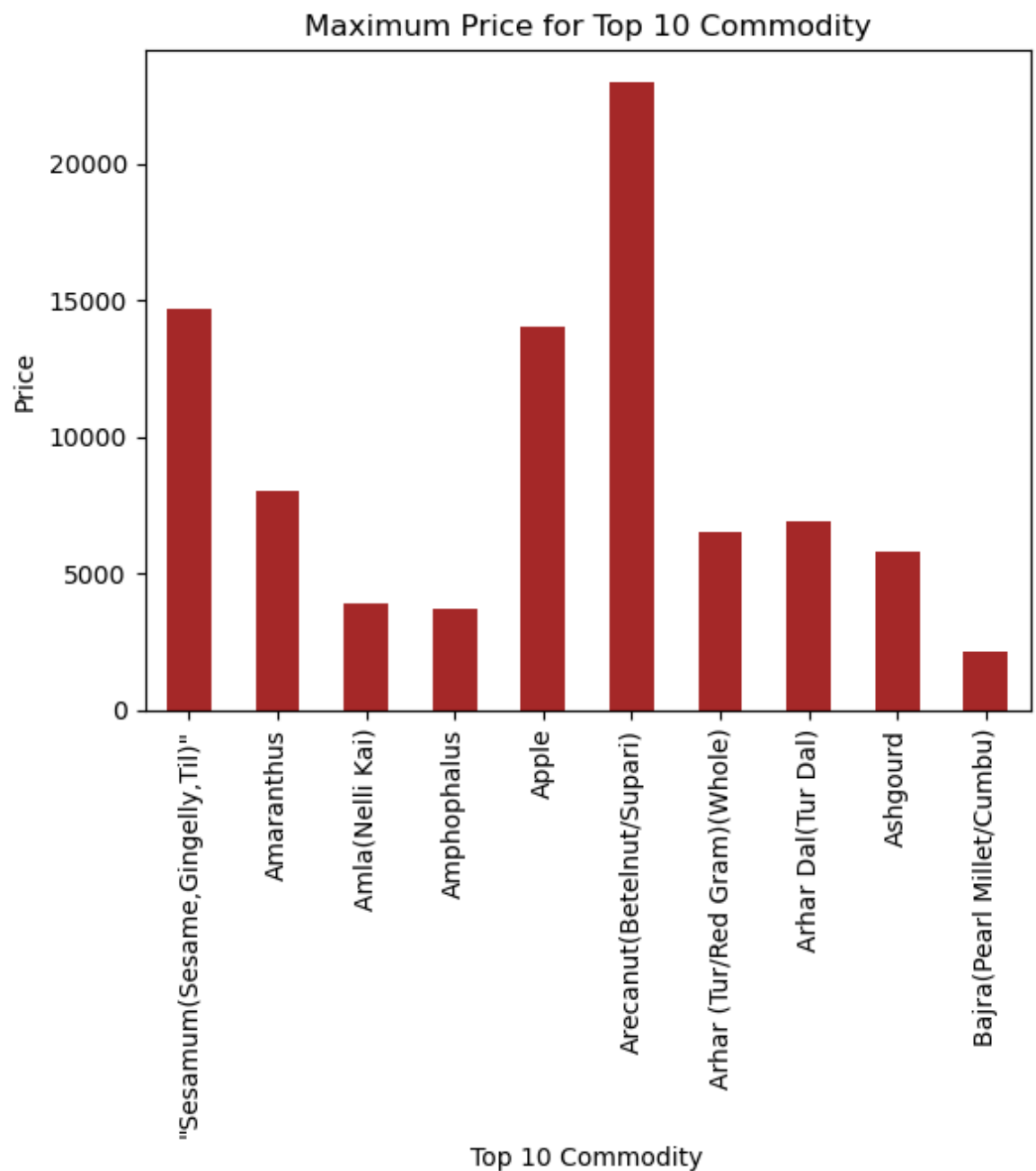


```
In [374]: ▶ #showing maximum price for top 10 commodity through barplots
max_prices = df.groupby('commodity').max()['max_price'][:10]

# Create a bar chart of the maximum prices
max_prices.plot(kind='bar', color = 'brown')

# Add a title and labels to the chart
plt.title('Maximum Price for Top 10 Commodity')
plt.xlabel('Top 10 Commodity')
plt.ylabel('Price')

# Show the chart
plt.show()
```

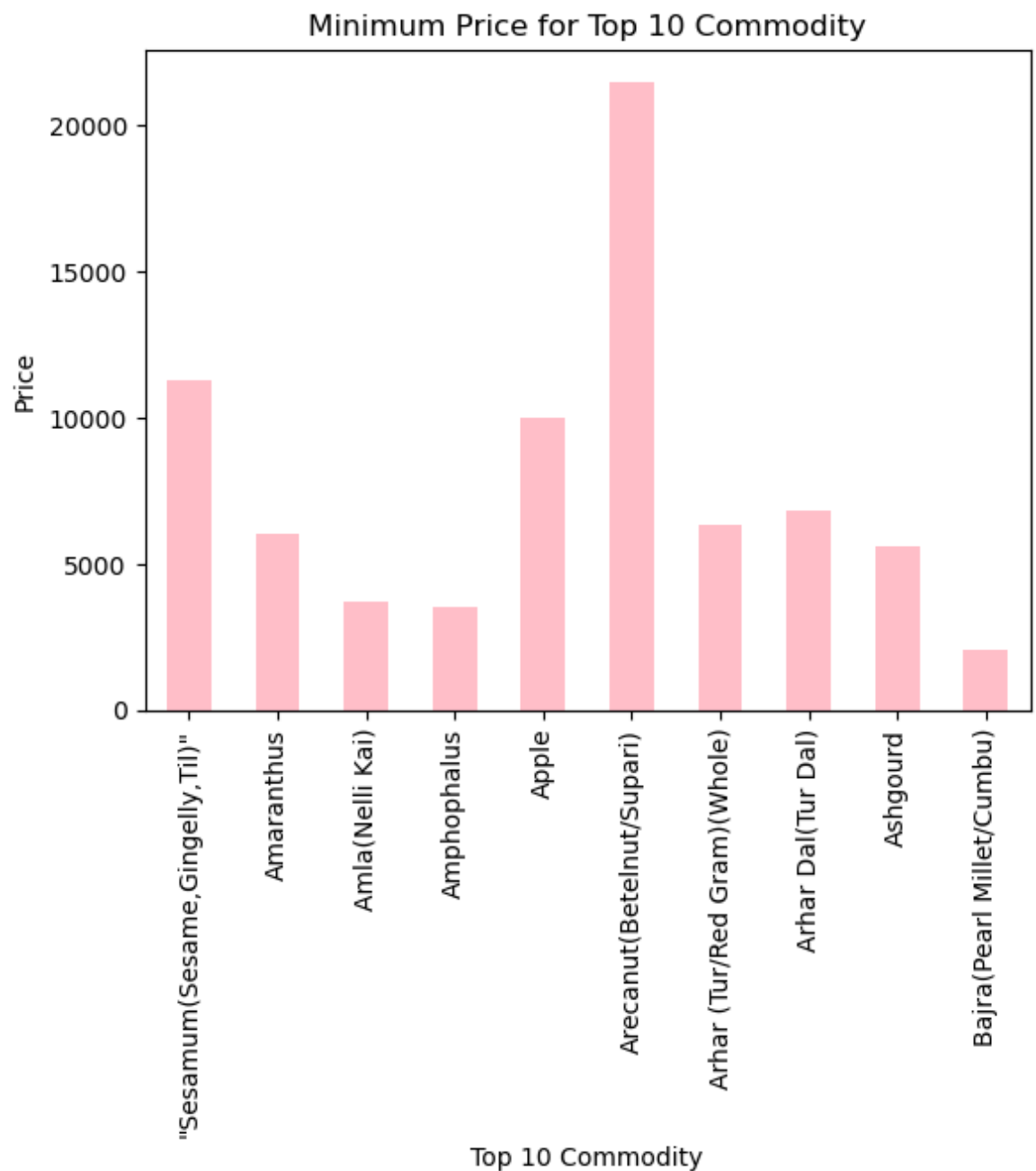


```
In [375]: ▶ #showing minimum price for top 10 commodity through barplots
min_prices = df.groupby('commodity').max()['min_price'][:10]

# Create a bar chart of the minimum prices
min_prices.plot(kind='bar', color = 'pink')

# Add a title and labels to the chart
plt.title('Minimum Price for Top 10 Commodity')
plt.xlabel('Top 10 Commodity')
plt.ylabel('Price')

# Show the chart
plt.show()
```

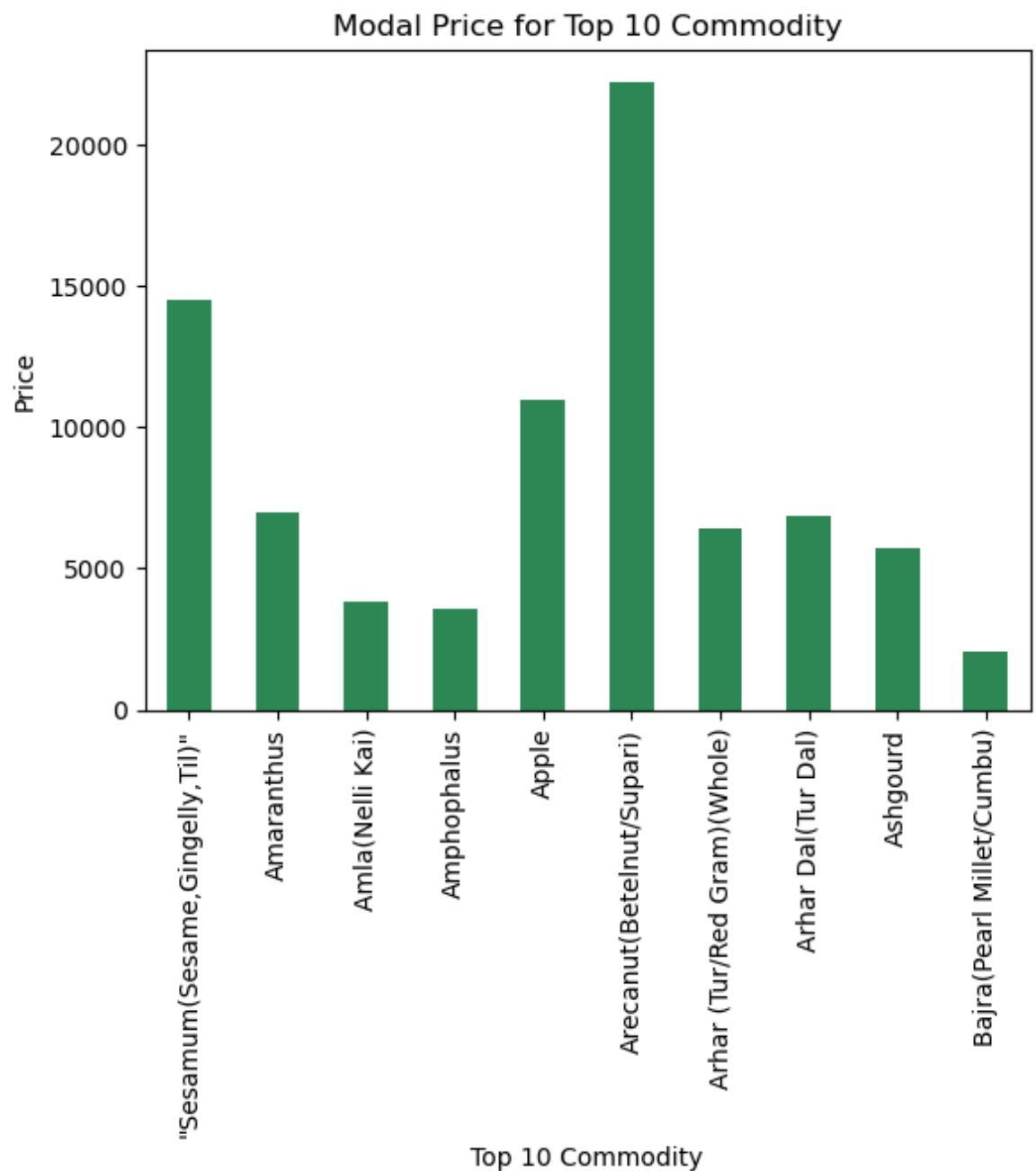


```
In [376]: #showing modal price for top 10 commodity through barplots
modal_prices = df.groupby('commodity').max()['modal_price'][:10]

# Create a bar chart of the minimum prices
modal_prices.plot(kind='bar', color = 'seagreen')

# Add a title and labels to the chart
plt.title('Modal Price for Top 10 Commodity')
plt.xlabel('Top 10 Commodity')
plt.ylabel('Price')

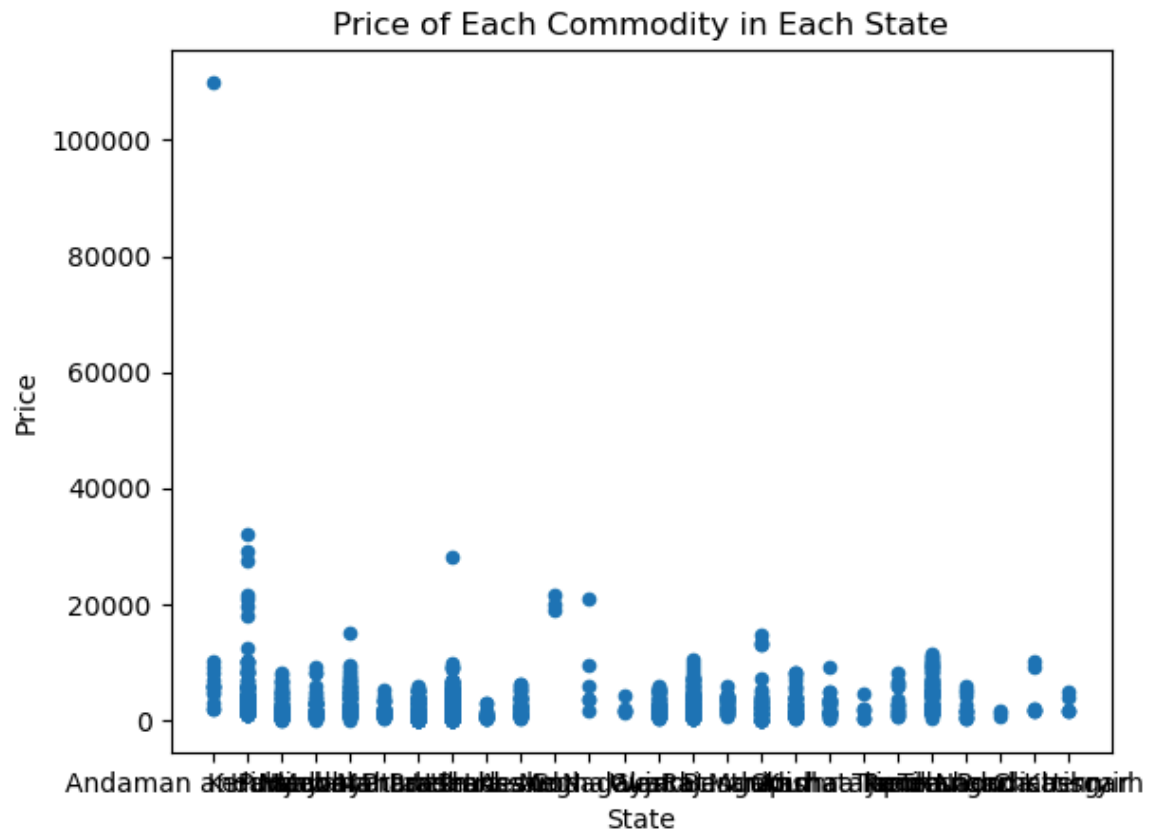
# Show the chart
plt.show()
```



```
In [377]: ▶ # showing scatter plot for the price of each commodity in each state
df.plot(kind='scatter', x='state', y='min_price')

# Add a title and labels to the chart
plt.title('Price of Each Commodity in Each State')
plt.xlabel('State')
plt.ylabel('Price')

# Show the chart
plt.show()
```



In [381]: `#showing the price of each commodity through catplots`

```
df = pd.read_csv("C:\\Users\\sd\\Downloads\\SHREYA KISPOTTA 22122317 - dataset\\data.csv")

# Create a catplot of the price of each commodity in each state
sns.catplot(x="state", y="min_price", hue="commodity", data=df, kind="point")

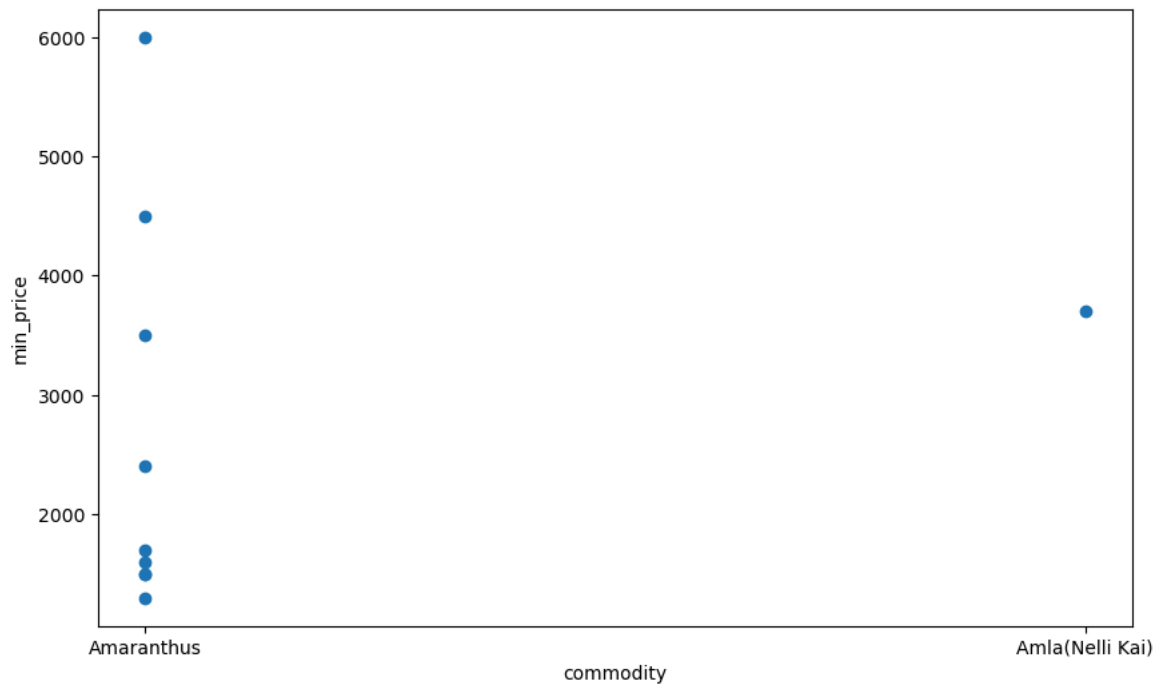
# Add a title and labels to the chart
plt.title('Price of Each Commodity in Each State')
plt.xlabel('State')
plt.ylabel('Price')

# Show the chart
plt.show()
```

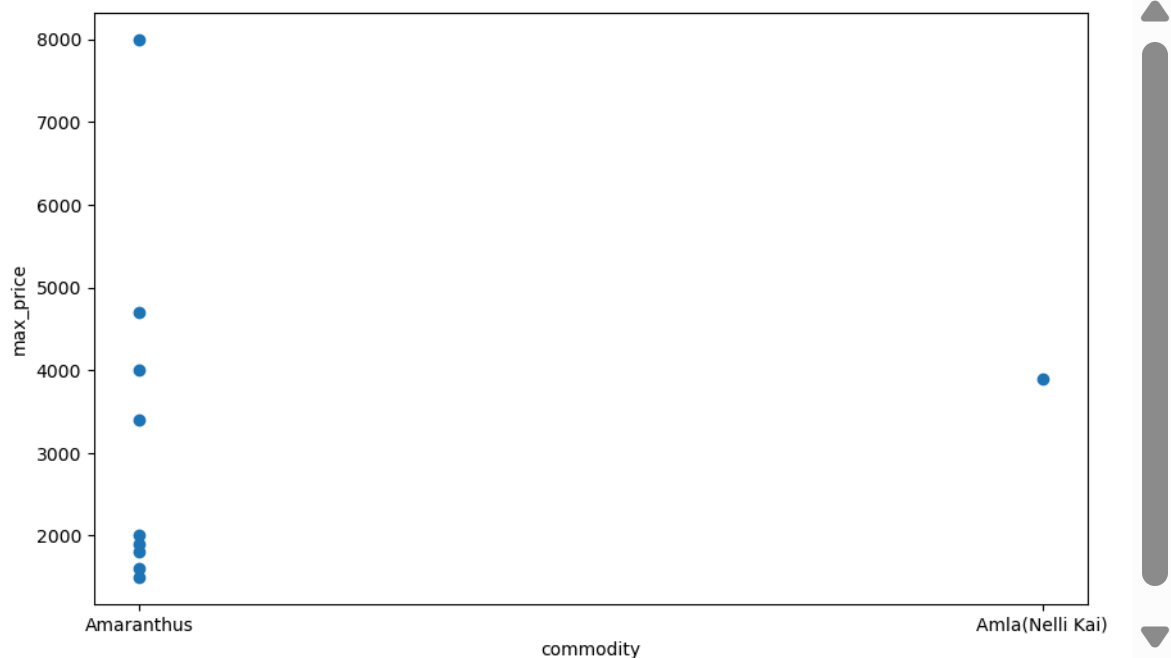
commodity

- Amaranthus
- Amla(Nelli Kai)
- Amphophalus
- Apple
- Arecanut(Betelnut/Supari)
- Arhar (Tur/Red Gram)(Whole)
- Arhar Dal(Tur Dal)
- Ashgourd
- Bajra(Pearl Millet/Cumbu)
- Banana
- Banana - Green
- Barley (Jau)
- Beans
- Beetroot
- Bengal Gram Dal (Chana Dal)
- Bengal Gram(Gram)(Whole)
- Ber(Zizyphus/Borehannu)
- Betal Leaves
- Bhindi(Ladies Finger)

```
In [383]: ▶ #showing the minimum price of each commodity
fig, ax = plt.subplots(figsize=(10,6))
ax.scatter(df['commodity'][:10], df['min_price'][:10])
ax.set_xlabel('commodity')
ax.set_ylabel('min_price')
plt.show()
```



```
In [387]: ▶ #showing the maximum price of each commodity
fig, ax = plt.subplots(figsize=(10,6))
ax.scatter(df['commodity'][:10], df['max_price'][:10])
ax.set_xlabel('commodity')
ax.set_ylabel('max_price')
plt.show()
```



```
In [389]: ▶ #showing the modal price of each commodity  
fig, ax = plt.subplots(figsize=(10,6))  
ax.scatter(df['commodity'][:10], df['modal_price'][:10])  
ax.set_xlabel('commodity')  
ax.set_ylabel('modal_price')  
plt.show()
```

